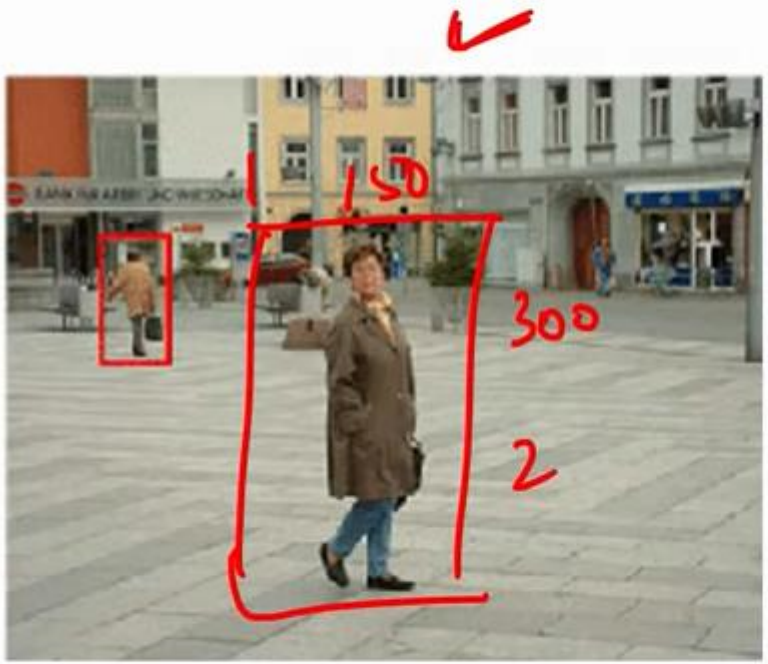
# Histogram of Oriented Gradients (HOG)

# HOG






150x300



64x128

8

8



8x16

| | | |
|---|---|---|
| 30 | 20 | 10 |
| 40 | 60 | 70 |
| 50 | 70 | 60 |

8 x 8

# HOG



8
x
8

| | | |
|---|---|---|
| 30 | 20 | 10 |
| 40 | 60 | 70 |
| 50 | 70 | 60 |

Grad Magnitude

| | | |
|---|---|---|
| | 20 | |
| 40 | 60 | 70 |
| | 70 | |

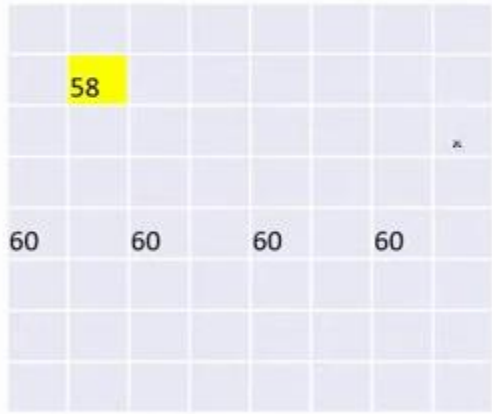
Grad Direction

X direction = |40 – 70| = 30
Y direction = |20 – 70| = 50

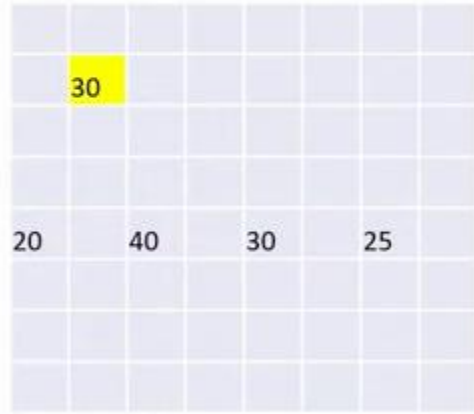
Grad Mag = sqrt(30^2 + 50^2) = ~58
Grad Direction = $\tan^{-1}(30/50) = \sim 30^0$

# HOG

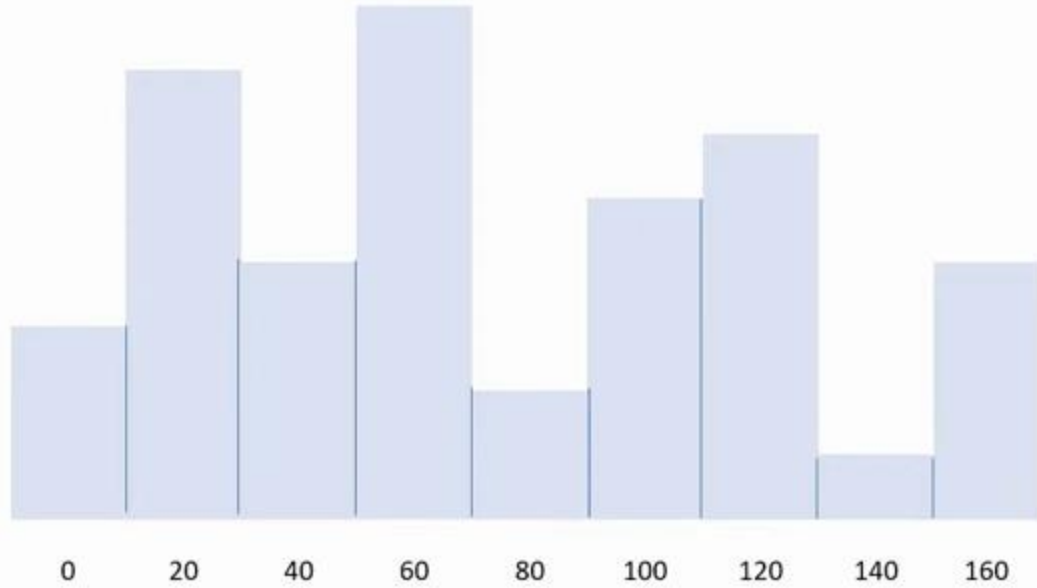


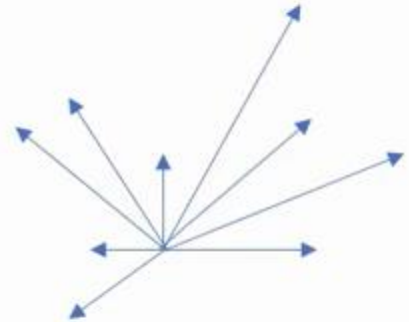



Grad Magnitude




Grad Direction



Feature Vector of size 9

# HOG



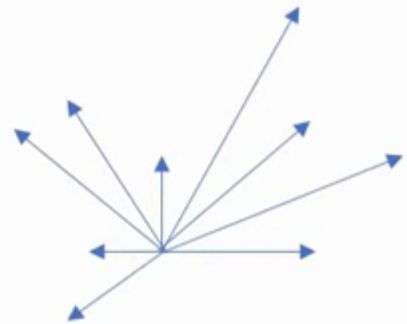
| 58 | | | |
|---|---|---|---|
| 60 | 60 | 60 | 60 |

Grad Magnitude

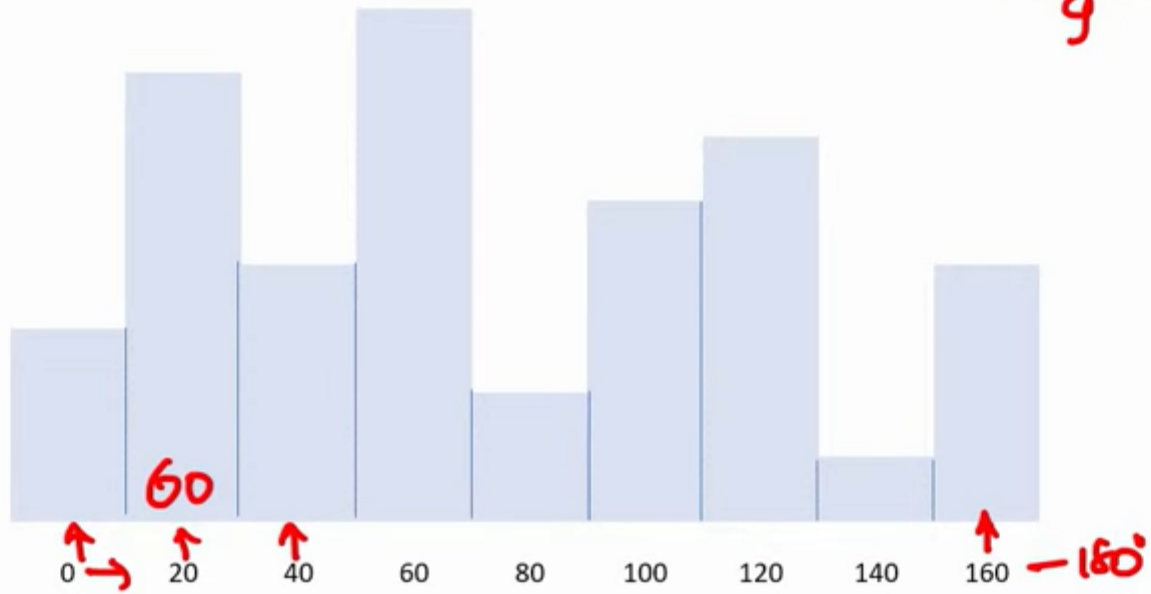
| 30 | | | |
|---|---|---|---|
| 20 | 40 | 30 | 25 |

Grad Direction

0 – 180°
9

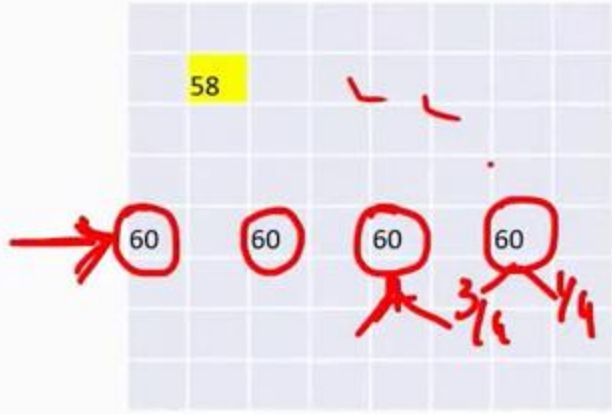


Feature Vector of size 9

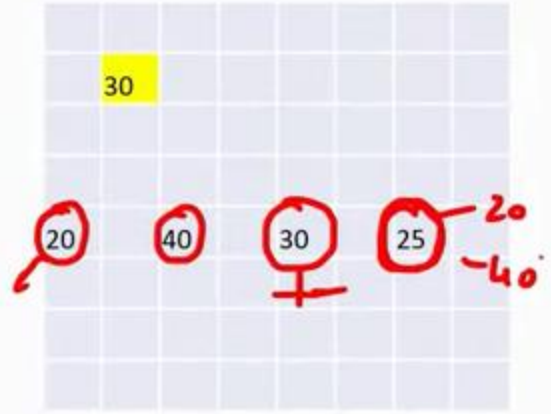


# HOG



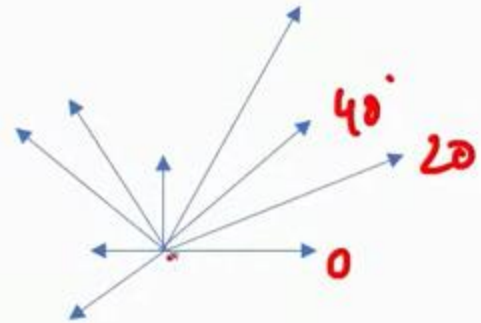


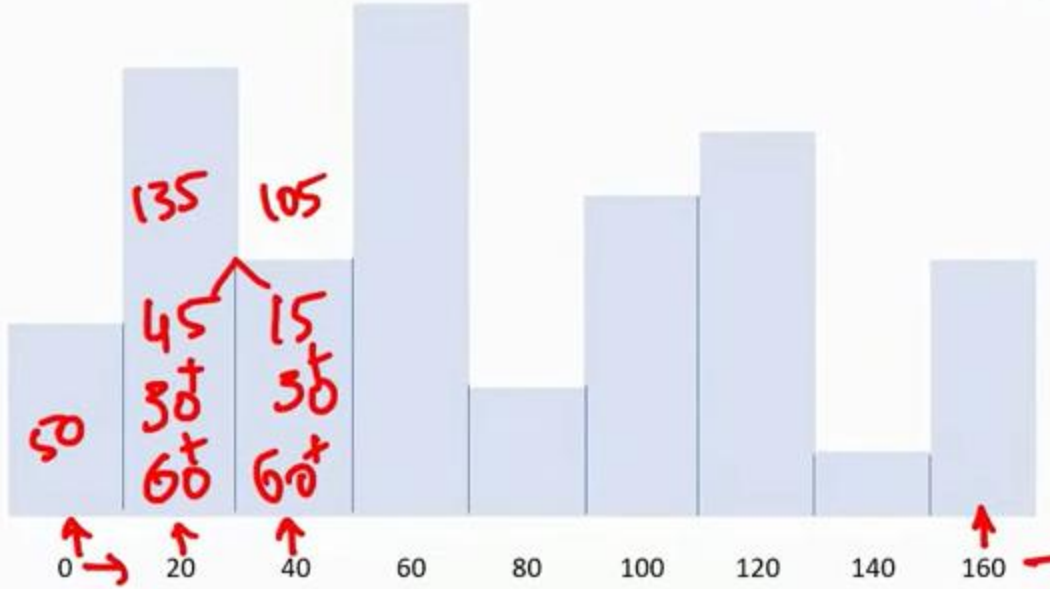

Grad Magnitude




Grad Direction

$\frac{0-180^\circ}{9}$




Feature Vector of size 9

→ Matrix $9 \times 1$

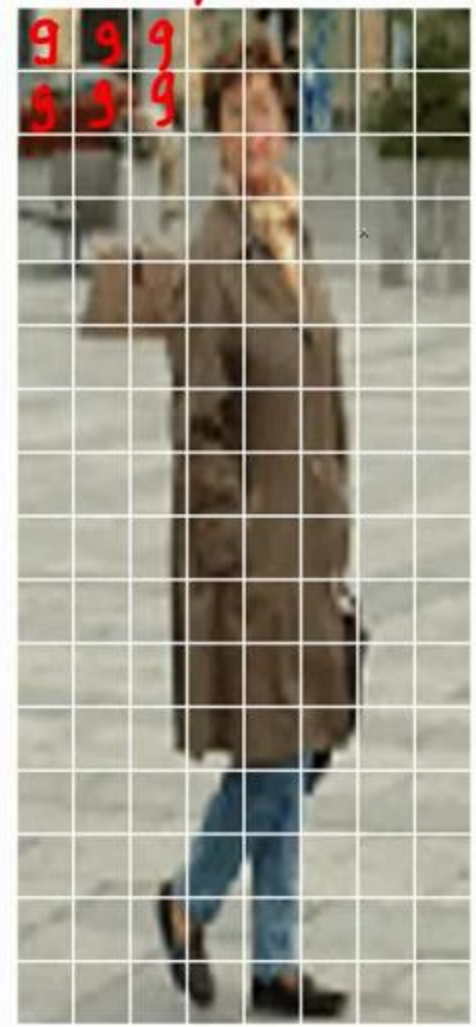
→ Vectors 9

FV = [ 50, 135, 105, - - - ] 9

# HOG Feature Vector




64x128



8x16

7*15*36 = 3780

# HOG Feature Vector




64x128




8x16

7x15

7*15*36 = 3780

150x300 = 45,000

2, 4 $\sqrt{2^2+4^2} = 4.472$

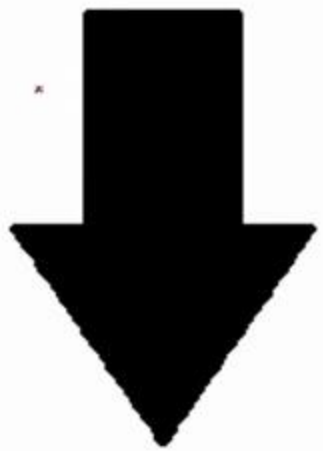
$\frac{2}{4.472}, \frac{4}{4.472} = 0.46, 0.89$

4, 8 $\sqrt{4^2+8^2} = 8.944$

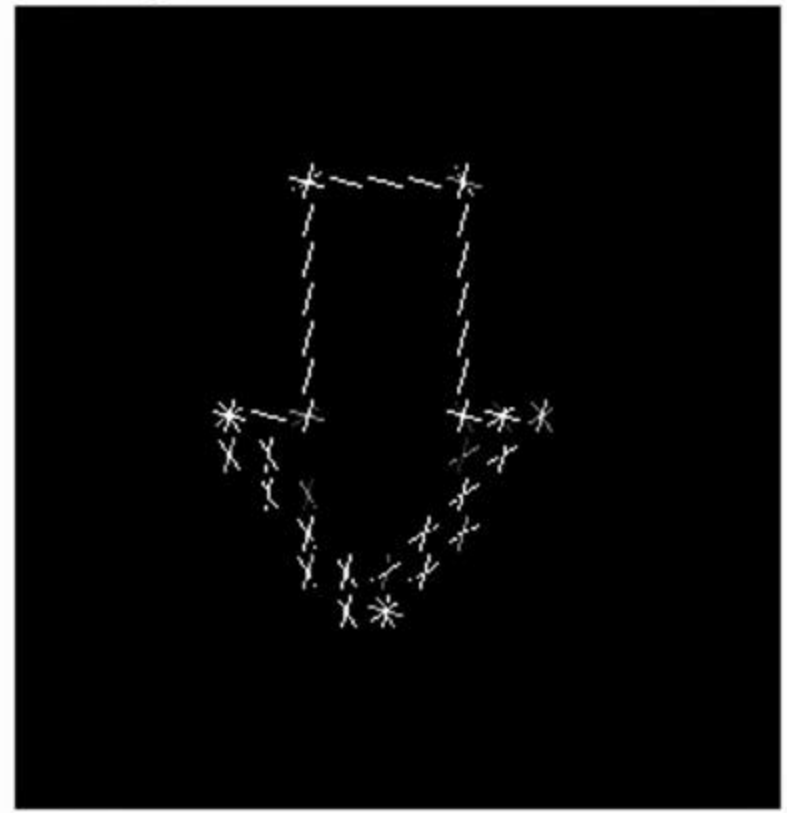
$\frac{4}{8.944}, \frac{8}{8.944} = 0.4472, 0.8944$

# HOG - Example

Input image



Histogram of Oriented Gradients



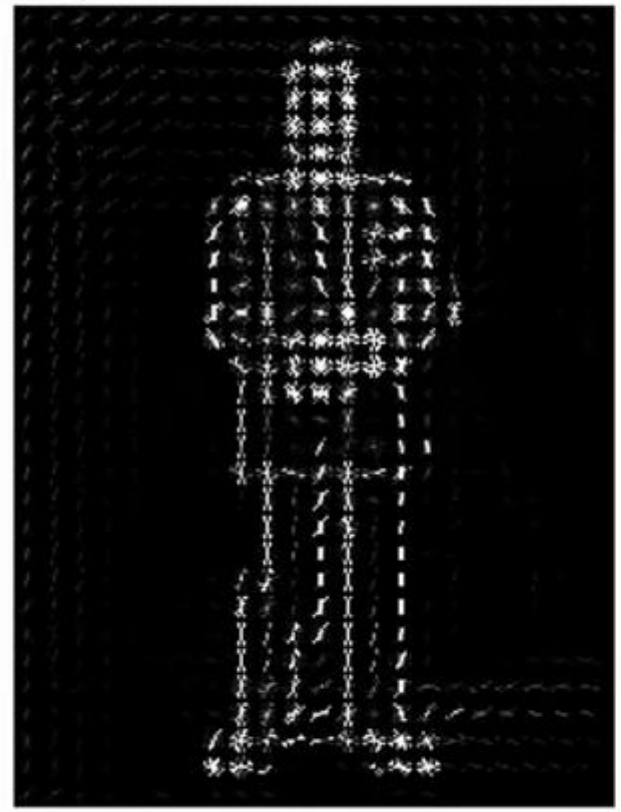


# HOG - Example

Input image



Histogram of Oriented Gradients





# Object Detection with HOG+SVM

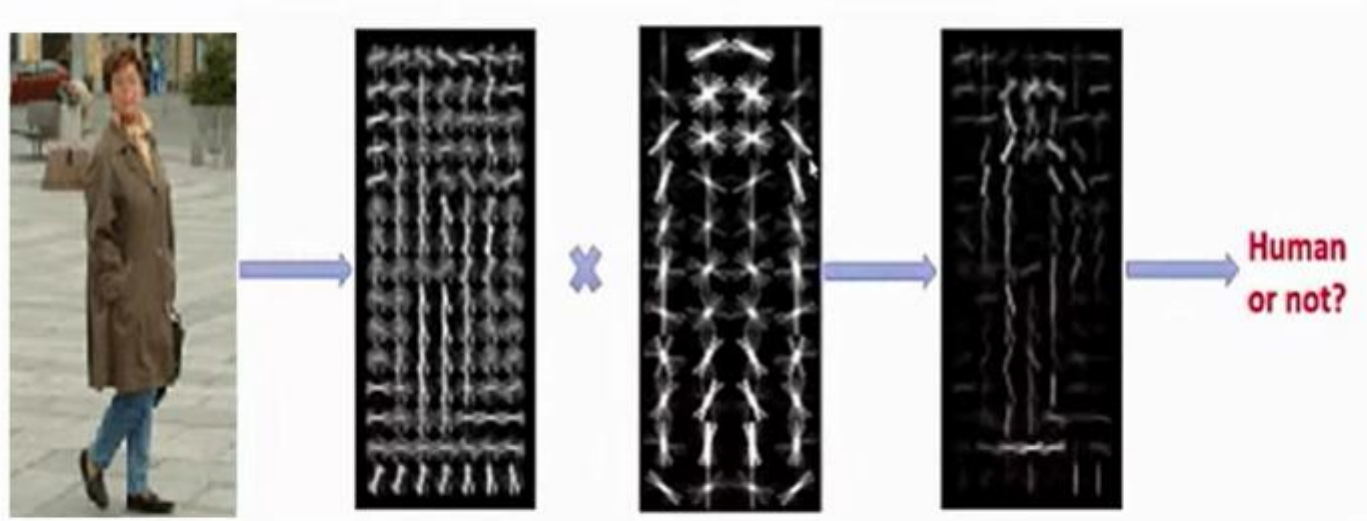


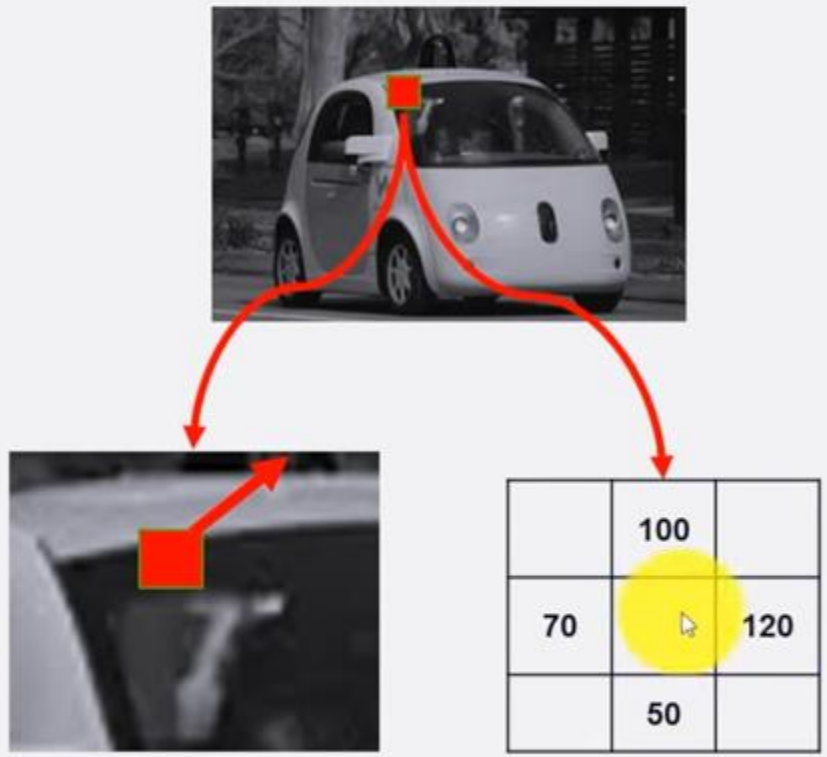

Image HOG Features Human Model Output of convolving with human model

Histograms of Oriented Gradients for Human Detection
Navneet Dalal and Bill Triggs

# HISTOGRAM OF ORIENTED GRADIENTS
## INTUITION

- The gradient value in the X-direction is 120-70=50
- Y-direction is 100-50=50.
- Putting it together we will have [50 50] feature vector.
- The magnitude and direction are calculated as follows:

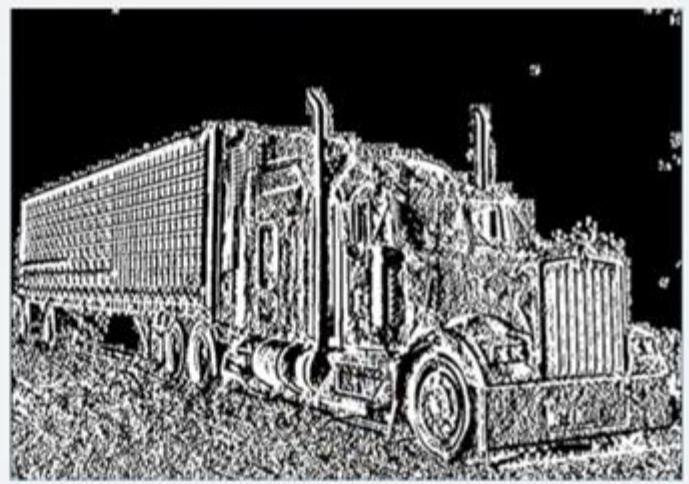
**Gradient Magnitude =** $\sqrt{(50)^2 + (50)^2} = 70.1$
**Gradient Angle =** $\tan^{-1}(50/50) = 45°$



# HISTOGRAM OF ORIENTED GRADIENTS
## STEPS



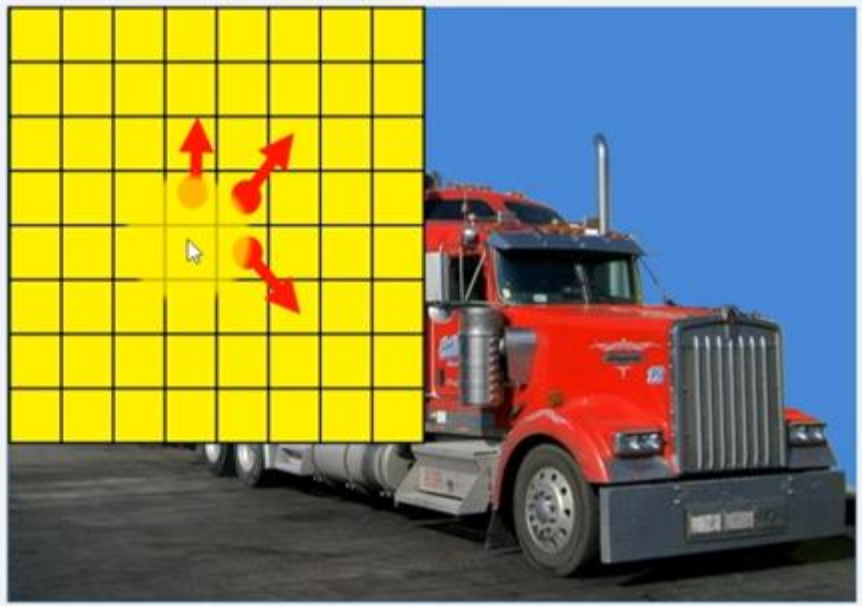




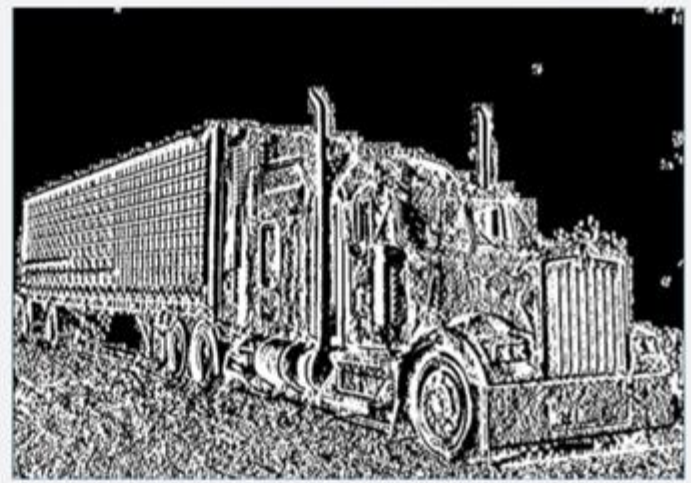
**Step 1:**

1. Using 8 x 8 pixel cell, compute gradient mag/direction

# HISTOGRAM OF ORIENTED GRADIENTS
## STEPS



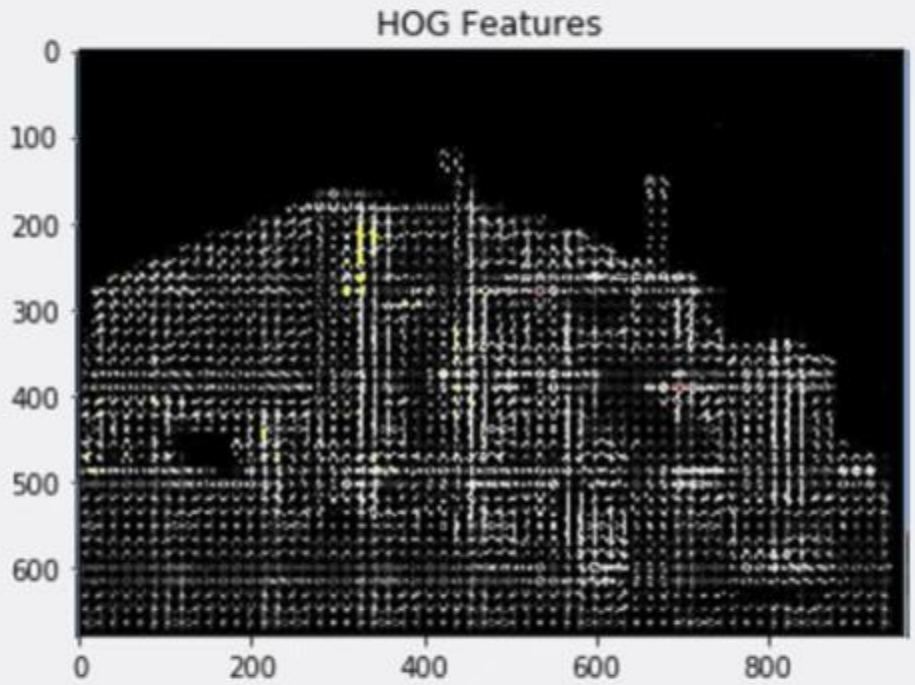


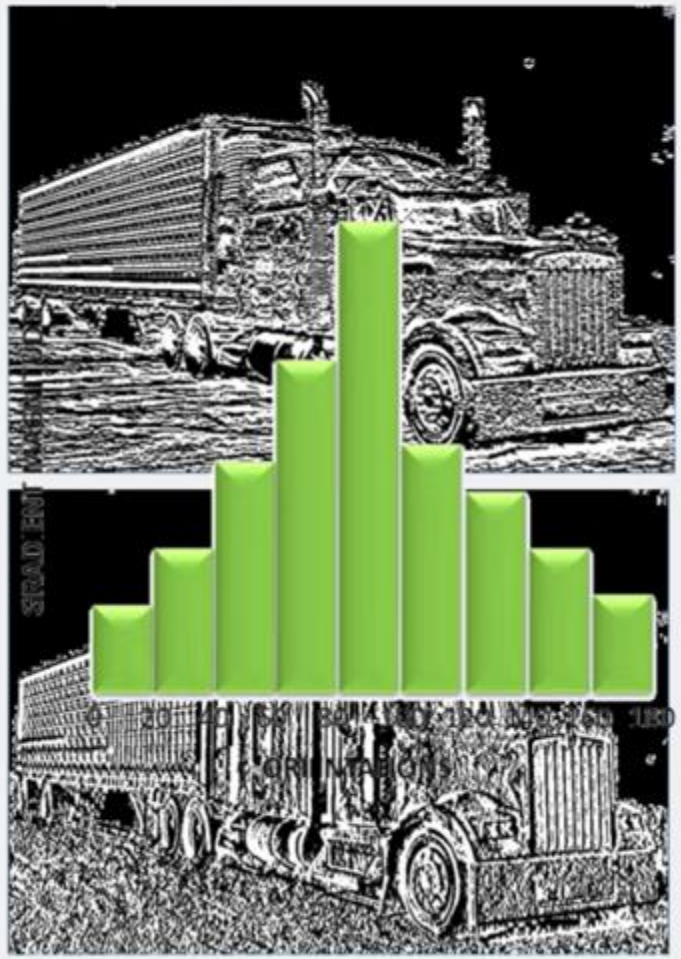


**Step 1:**

1. Using 8 x 8 pixel cell, compute gradient mag/direction

# HISTOGRAM OF ORIENTED GRADIENTS
## STEPS



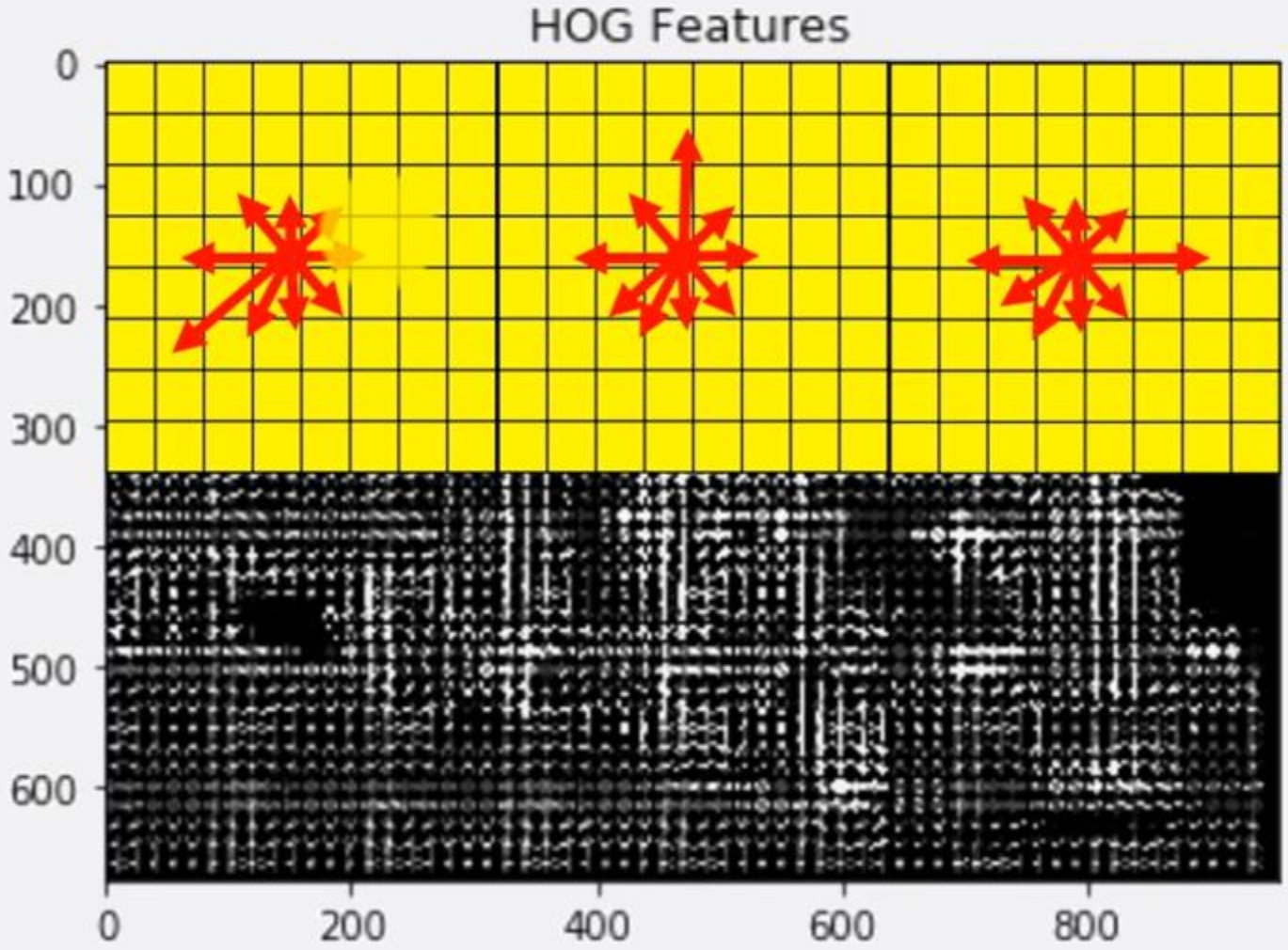
**Step 1:**

1. Using 8 x 8 pixel cell, compute gradient mag/direction

**Step 2:**

- Create a histogram of generated 64 (8 x 8) gradient vectors
- Each cell is then split into angular bins, each bin corresponds to a gradient direction (9 bins 0-180°) (20° each bin).
- This reduces 64 vectors to just 9 values.

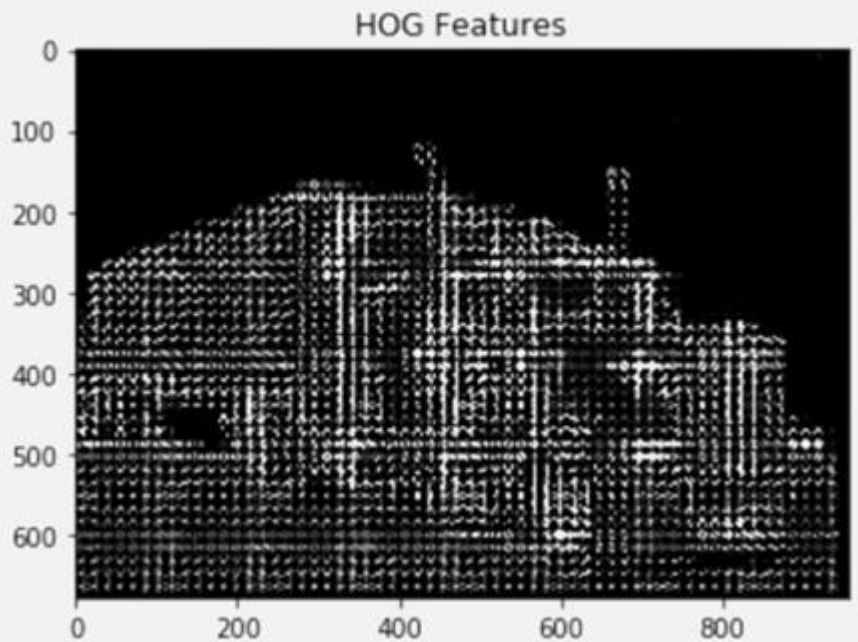
# HISTOGRAM OF ORIENTED GRADIENTS
## EXPECTED HOG OUTPUT



# HISTOGRAM OF ORIENTED GRADIENTS
## SUMMARY

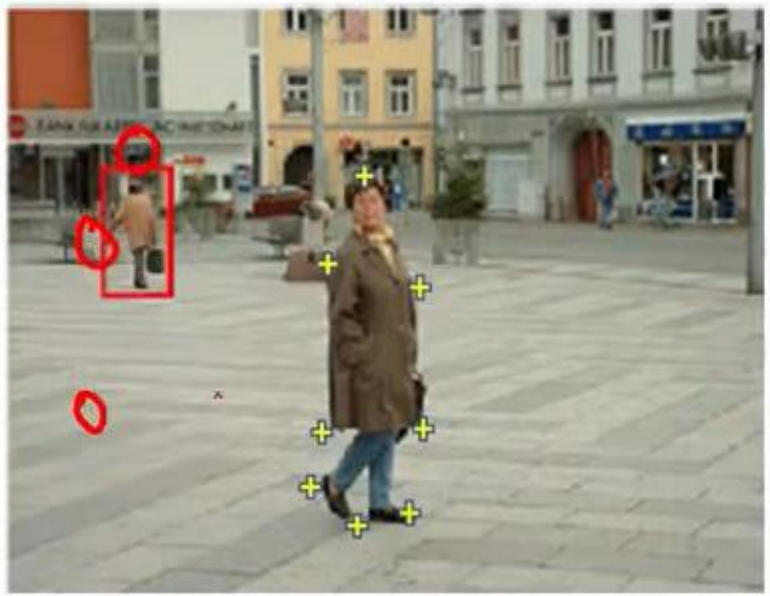
- HOGs are a feature descriptor used for object detection.
- HOGs along with SVM (support vector machine) classifiers work well for object detection.
- HOG applies a sliding window detector over an image and HOG descriptor is computed for each position.
- HOG takes care of the scaling issues by changing image scale (pyramiding).





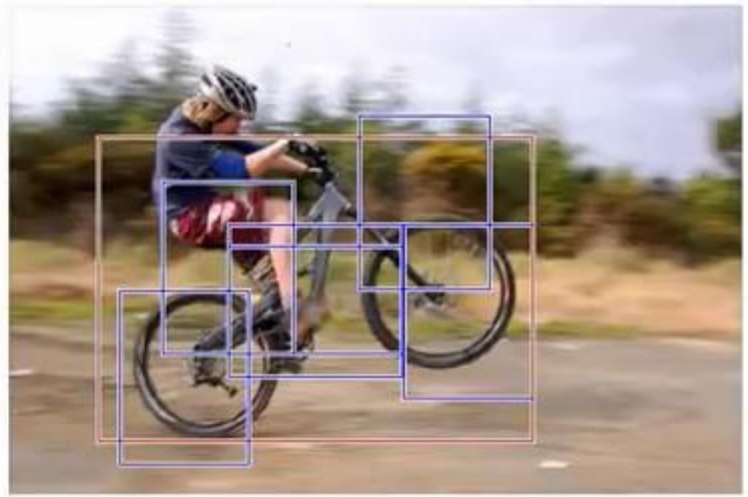
# SIFT vs HOG

- Why didn't I use Sliding Windows in SIFT?



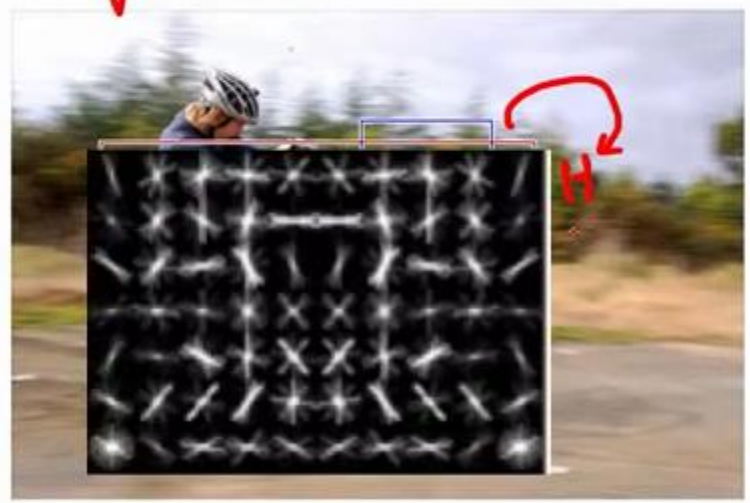
# SIFT vs HOG

- Why might the D&T detector fail to detect the wheelie-bicycle?



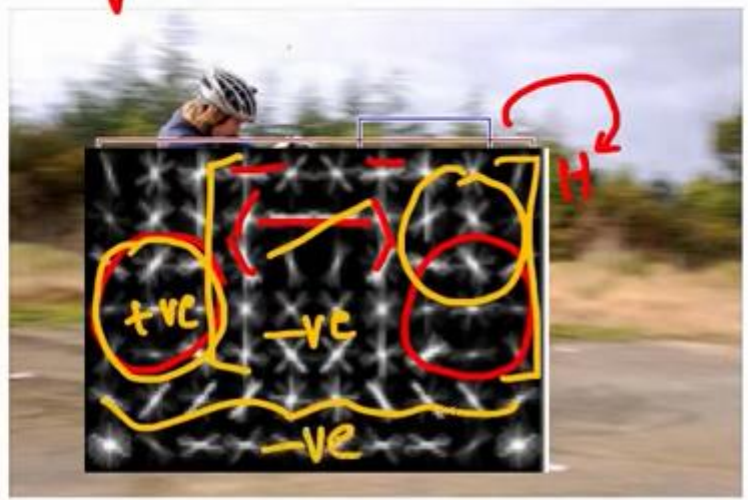
# SIFT vs HOG

- Why might the D&T detector fail to detect the wheelie-bicycle?



# SIFT vs HOG

- Why might the D&T detector fail to detect the wheelie-bicycle?



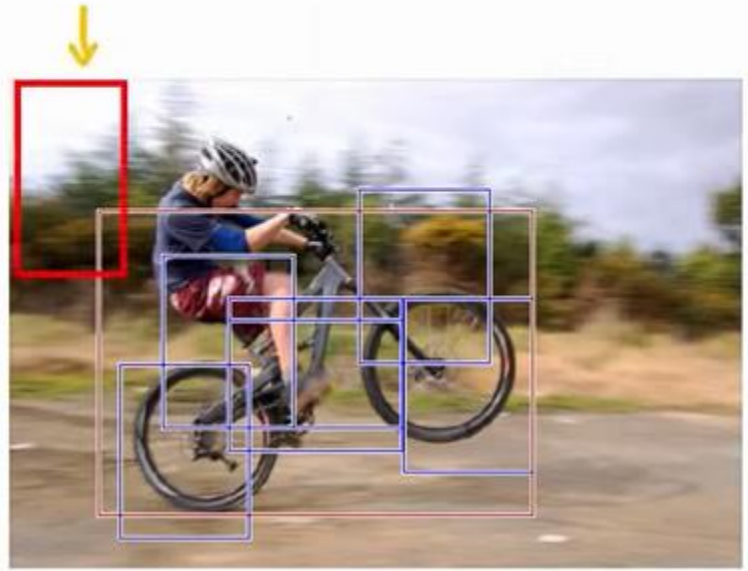
# SIFT vs HOG

- Why might the D&T detector fail to detect the wheelie-bicycle?



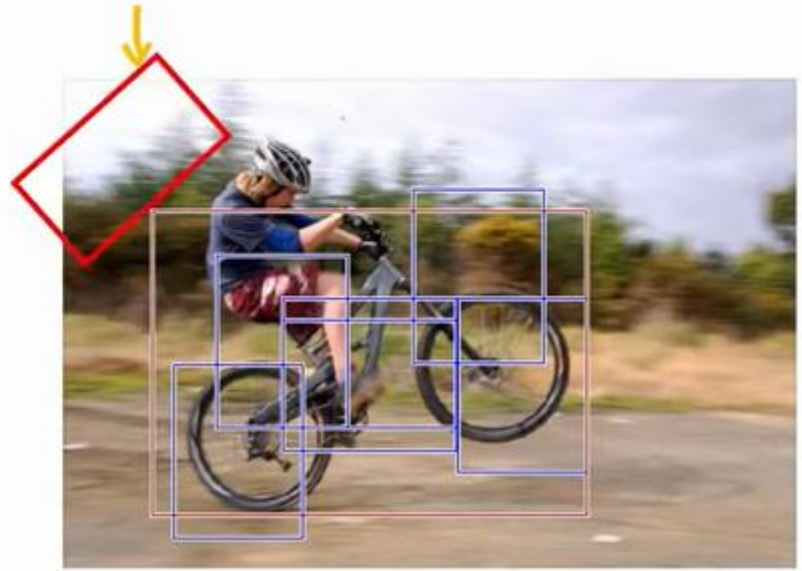
# SIFT vs HOG

- Why might the D&T detector fail to detect the wheelie-bicycle?



# SIFT vs HOG

- Why might the D&T detector fail to detect the wheelie-bicycle?



# SIFT vs HOG

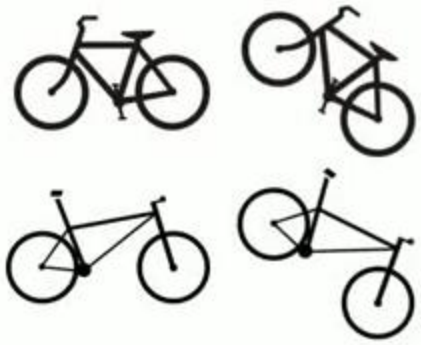
- Why might the D&T detector fail to detect the wheelie-bicycle?

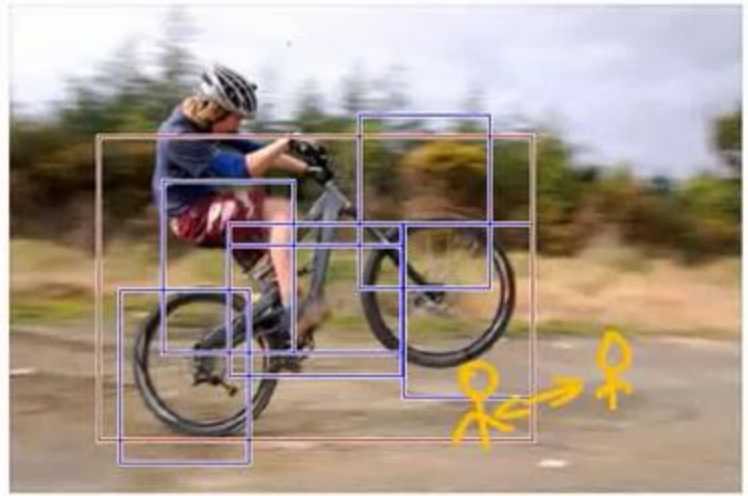




# SIFT vs HOG

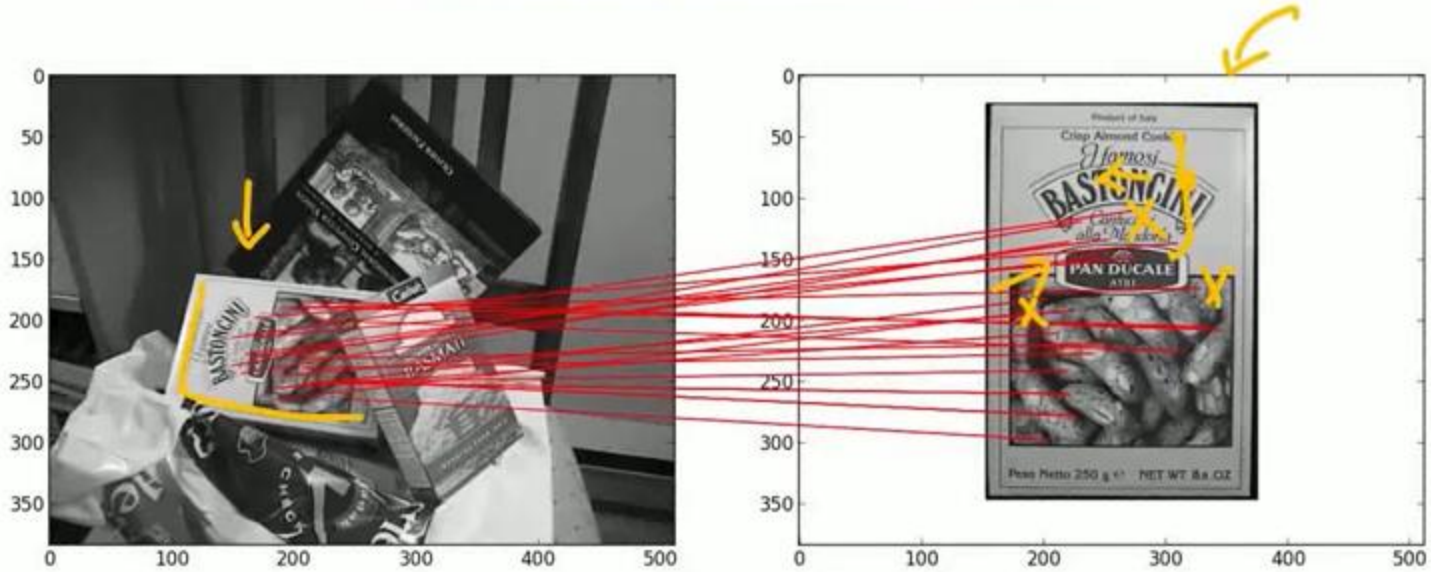
- Why might the D&T detector fail to detect the wheelie-bicycle?



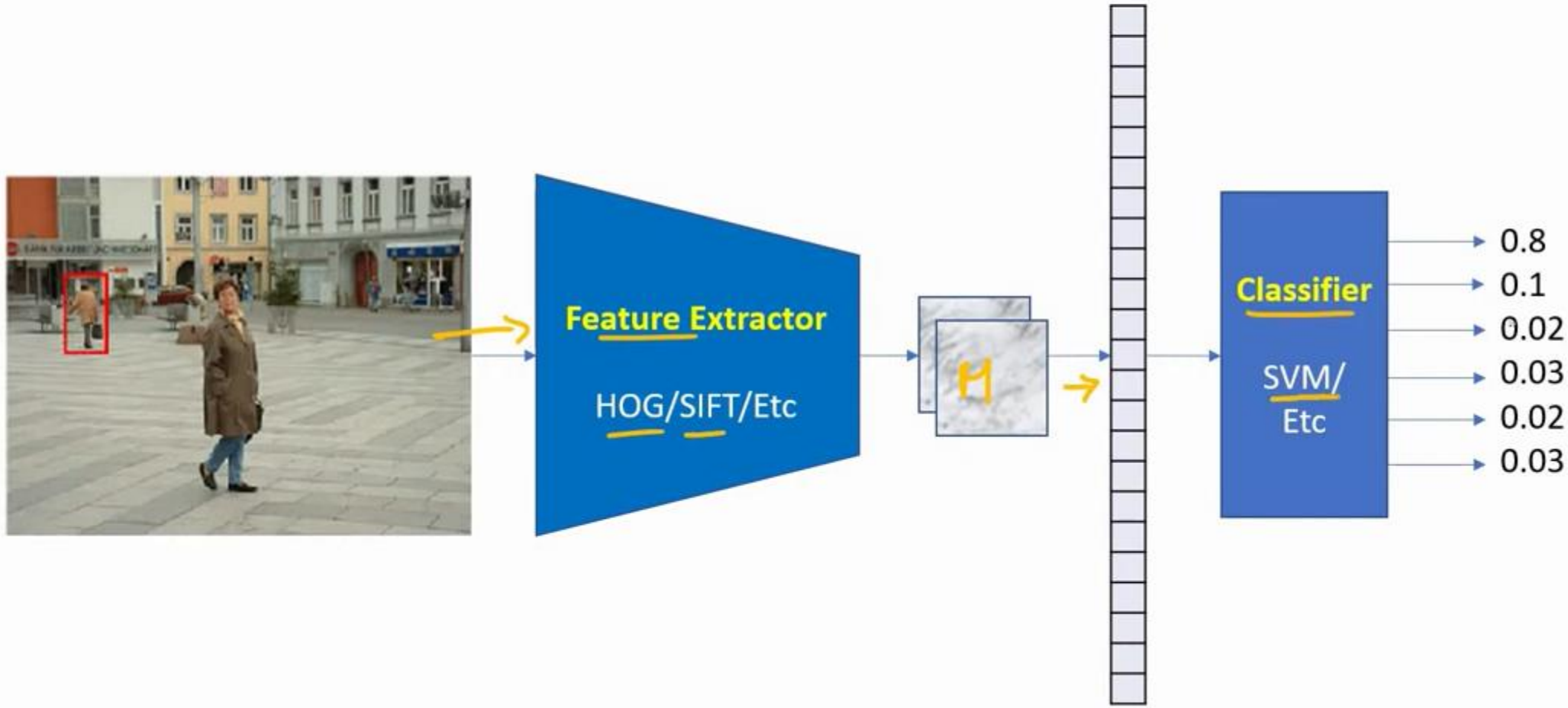


Feature Extraction

- Key/Interest Points
  - Rotation Invariance
  - Translation invariance
  - Illumination invariance
  - Scale invariance
- Dense Sampling



# Object Detection Pipeline



# Reference

- https://www.youtube.com/watch?v=28xk5i1_7Zc
- https://www.youtube.com/watch?v=XmO0CSsKg88
- https://www.youtube.com/watch?v=Z2ml7WzCrJ8